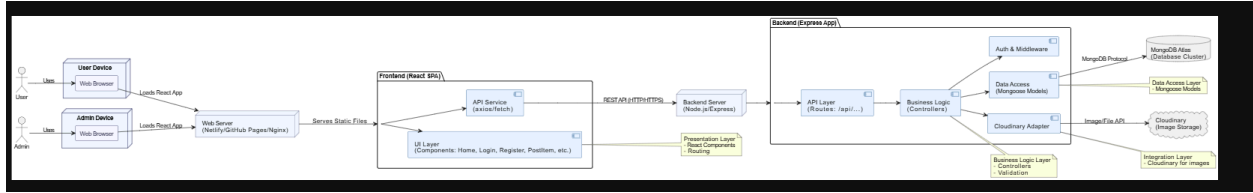


System Architecture

Architectural Diagram:

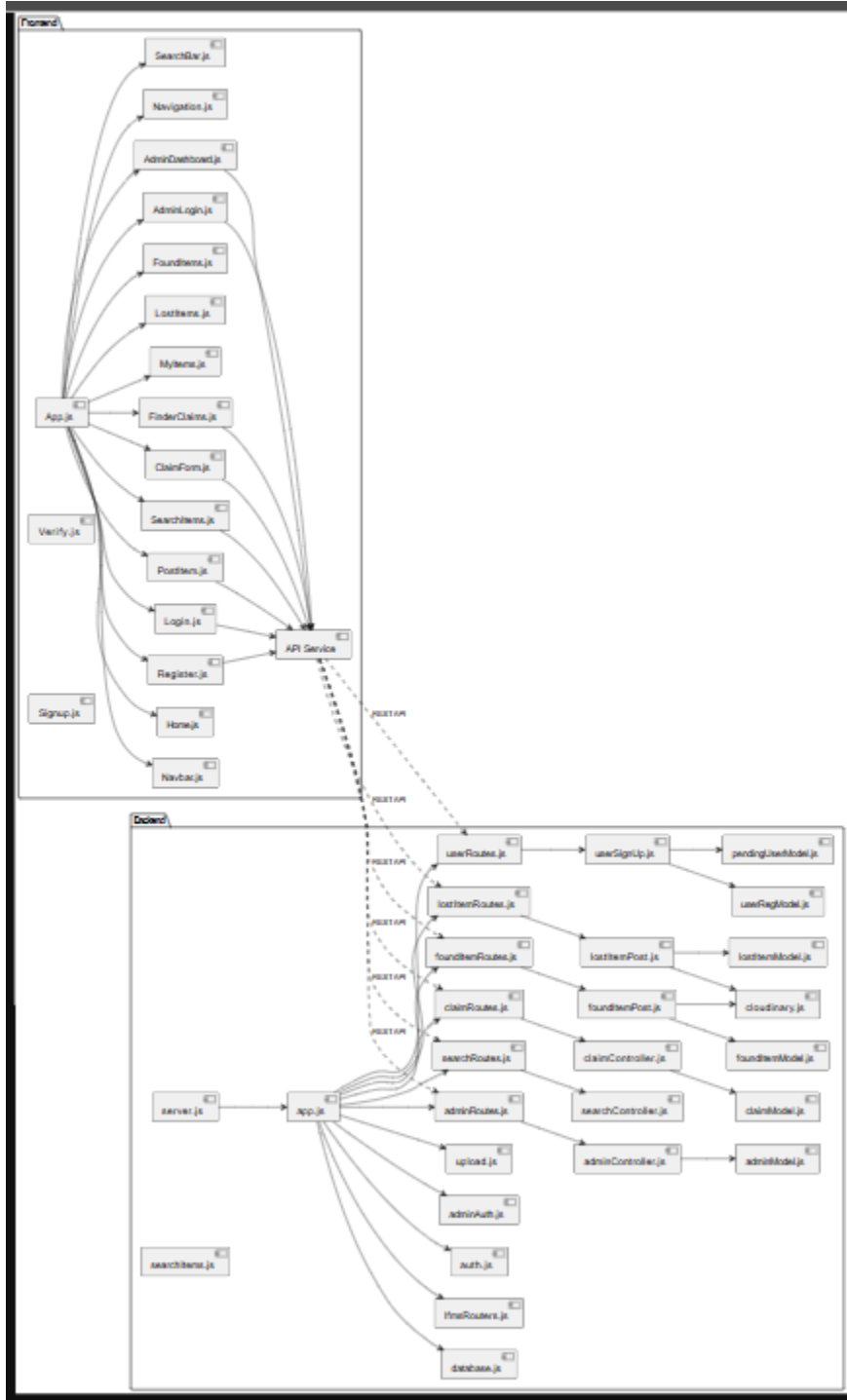


Architectural Styles Used

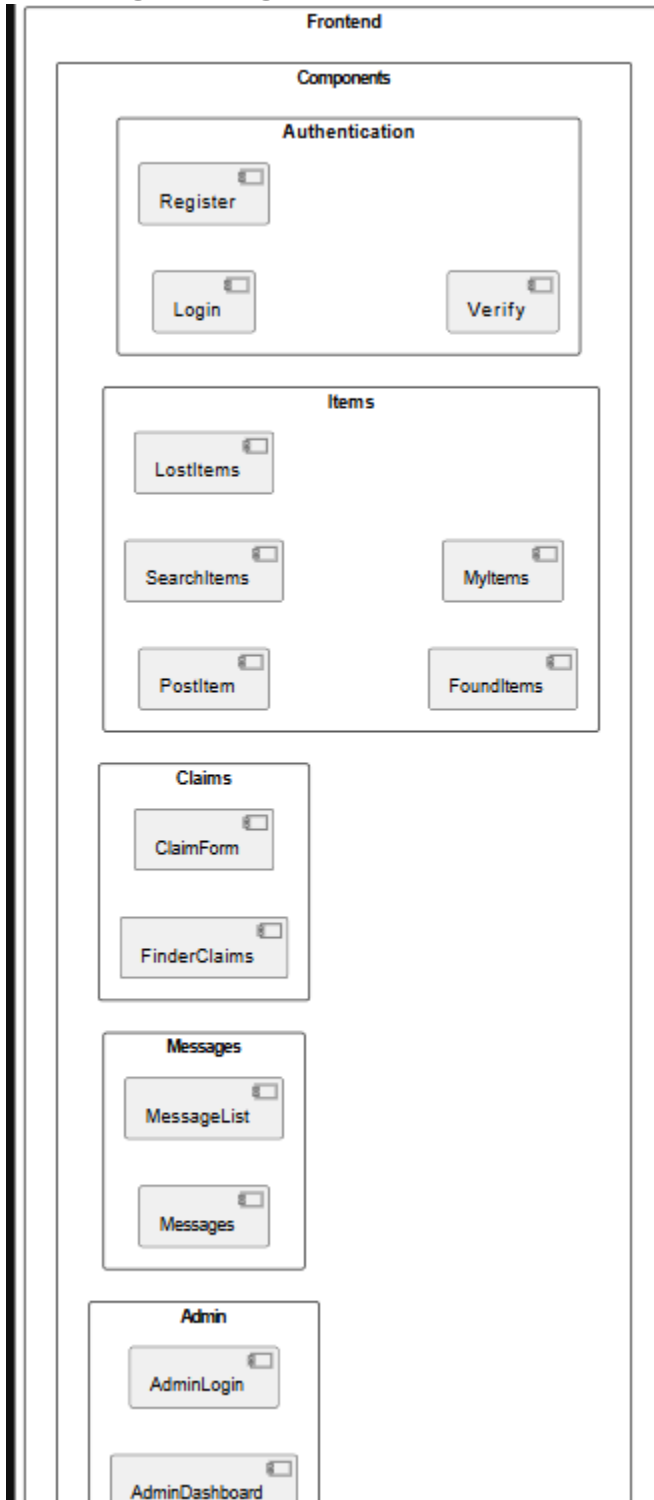
From the [ArchitecturalDiagram.drawio.html](#):

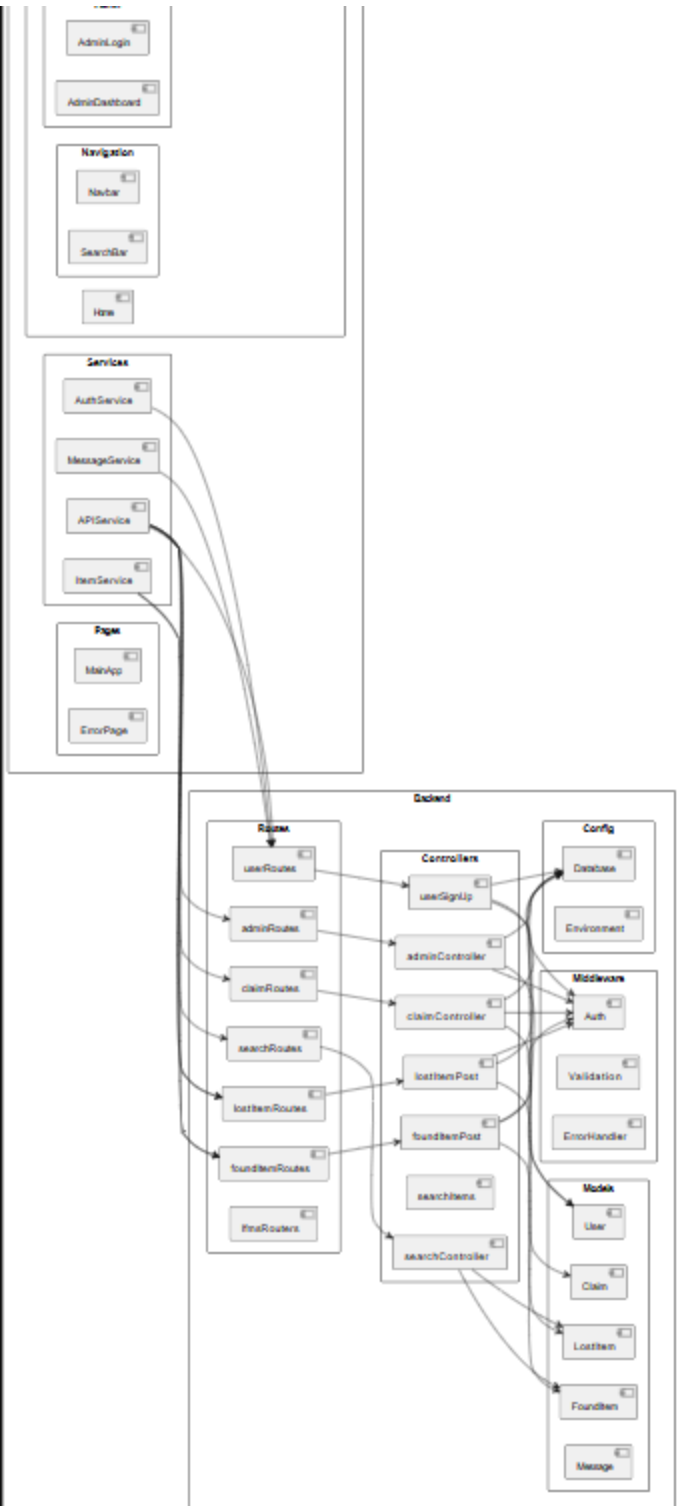
- The system uses:
 - **Layered Architecture:**
 - Frontend Layer (React SPA)
 - Backend Layer (Express App with Node.js)
 - Data Storage Layer (MongoDB Atlas)
 - External Service Layer (Cloudinary)
 - **Client-Server Architecture:**
 - Clients: User and Admin devices (Web browsers)
 - Servers: Web Server (Netlify/GitHub Pages/Nginx), Backend Server (Node.js/Express)
 - **Deployment Context:**
 - Web Browser loads React App from Web Server
 - React App communicates with Backend Server using REST APIs (HTTP/HTTPS)
 - Backend Server interacts with MongoDB Atlas (database) and Cloudinary (for file storage)

Component Diagram:



Package Diagram:





UML Package Diagram of Subsystems

- **Frontend:**
 - **Components:**
 - **Authentication:**
 - Login
 - Register
 - Verify
 - **Items:**
 - PostItem
 - SearchItems
 - LostItems
 - FoundItems
 - MyItems
 - **Claims:**
 - ClaimForm
 - FinderClaims
 - **Messages:**
 - MessageList
 - Messages
 - **Admin:**
 - AdminLogin
 - AdminDashboard
 - **Navigation:**
 - Navbar
 - SearchBar

- Home
- **Services:**
 - ApiService
 - AuthService
 - ItemService
 - MessageService
- **Pages:**
 - MainApp
 - ErrorPage
- **Backend:**
 - **Controllers:**
 - userSignUp
 - lostItemPost
 - foundItemPost
 - claimController
 - searchController
 - searchItems
 - adminController
 - **Routes:**
 - userRoutes
 - lostItemRoutes
 - foundItemRoutes
 - claimRoutes
 - searchRoutes
 - adminRoutes

- IfmsRouters

- **Models:**

- User
- LostItem
- FoundItem
- Claim
- Message

- **Middleware:**

- Auth
- Validation
- ErrorHandler

- **Config:**

- Database
- Environment

- **Dependencies:**

- Frontend services (APIService, AuthService, ItemService, MessageService) depend on backend routes (userRoutes, lostItemRoutes, foundItemRoutes, claimRoutes, searchRoutes, adminRoutes).
- Backend routes depend on corresponding backend controllers.
- Backend controllers depend on models, middleware, and configuration settings.

